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# =====
# Iris Flower Classification
# =====
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
# =====
# 1. Load Dataset
# =====
iris = load_iris()
df = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)
df['species'] = iris.target
print("First 5 rows:")
print(df.head())
# =====
# 2. Data Exploration
# =====
print("\nDataset Information:")
print(df.info())
print("\nDataset Description:")
print(df.describe())
# =====
# 3. Split Data into Training and Testing
# =====
X = df[iris.feature_names]
y = df['species']
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
# =====
# 4. Train Machine Learning Model
# =====
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
# =====
# 5. Prediction
# =====
y_pred = model.predict(X_test)
print("\nPredicted Values:")
print(y_pred)
# =====
# 6. Model Evaluation
# =====
accuracy = accuracy_score(y_test, y_pred)
print("\nModel Accuracy:")
print(accuracy * 100, "%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d"
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
# =====
# 7. Data Visualization
# =====
sns.scatterplot(
    data=df,
    x="sepal length (cm)",
    y="sepal width (cm)",
    hue="species"
)

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)  
plt.title("Iris Flower Classification")  
plt.show()
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